STUDENT LAB WORKBOOK • UNITS 1–12

HANDS-ON MATH LAB

Student Explorations from Physical Blocks to the Stack



StackCalc

STUDENT WELCOME & QUICK-START GUIDE

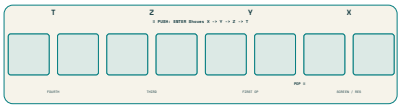
Welcome to the Hands-On Math Lab!

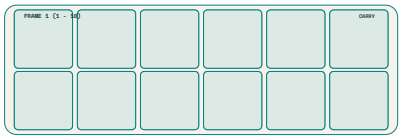
How to explore every math concept with physical blocks, sketches, and your StackCalc RPN calculator.

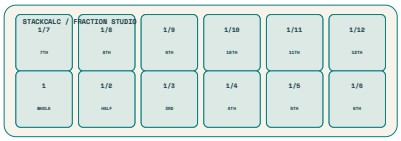
THE GOLDEN RULE OF THE MATH LAB: BUILD IT, SKETCH IT, PREDICT IT, CHECK IT!

In this workbook, your calculator is never a black box that hides the math. Every unit starts with a real-world story and a physical block challenge. You will build the problem with your hands-on blocks first, draw a quick sketch, write down your predicted answer, and then press the keys on your StackCalc calculator to watch the 4-level stack verify your thinking!

HOW THE 4-LEVEL STACK ELEVATOR (T, Z, Y, X) WORKS







MEET THE FOUR FLOORS OF YOUR CALCULATOR'S MEMORY

- Floor 1 — Register X (Active Screen): Shows the number you are typing right now or the answer just calculated.
- Floor 2 — Register Y (Waiting Shelf): Holds your first number after you press [ENTER] while you type the second number.
- Floors 3 & 4 — Registers Z & T (Upper Shelves): Hold earlier pieces of a multi-step problem—no parentheses needed!
- **Try Your First Stack Calculation (8 + 5 = 13):** 8 ENTER 5 +
- **Try a Two-Dot Fraction (1/2 + 1/4 = 3/4):** • 1 • 2 ENTER • 1 • 4 +

Checking Your Work with the Self-Check Answer Key

After you finish the 4 investigation problems in any unit, flip to the Student Self-Check Answer Key on Page 25 to verify your answers and keystroke sequences!

STUDENT TABLE OF CONTENTS & PROGRESS LOG

Your 12 Hands-On Math Lab Units

Check off each unit as you complete the Guided Walkthrough and the 4-Problem Lab Investigation.

Unit & Mathematical Topic	Grades & Calculator	Companion Book Story	Pages / Done
Unit 01: Counting, Cardinality & The One- Kit: Ten-Frame Tray & Counting Pegs	Pre-K – Kind (SC-6)	How Ten Fingers & an Empty	pp. 1–2 [ ]
Unit 02: Part-Part-Whole & Doubles: The M Kit: Math Domino Studio (12-Frame Tray) + D	Grades K–1 (SC-6)	Spotting Number Partners a	pp. 3–4 [ ]
Unit 03: Ten-Frames: Complements to 10 & Kit: Two 10-Frame Trays (2x5 each, 10 slots	Grade 1 (SC-6)	Making a Ten First (8 + 5	pp. 5–6 [ ]
Unit 04: Number Racks (Rekenrek): 5-Struc Kit: Two-Row Number Rack (Rekenrek)	Grades 1–2 (SC-6)	Two Hands, One Workbench:	pp. 7–8 [ ]
Unit 05: The 4-Register Stack: Four Opera Kit: Place Value Blocks & 4-Register Stack	Grades 3–4 (SC-6)	Inside the 4-Floor Number	pp. 9–10 [ ]
Unit 06: Fraction Studio: Height Equivale Kit: 84 mm Fraction Towers & Twin Comparato	Grades 4–5 (SC-6)	The Pharaoh's Bakery, Pyth	pp. 11–12 [ ]
Unit 07: Postfix Evaluation & Chained Ope Kit: Counting & Array Tiles + 4-Register St	Grades 5–6 (SC-6)	Why 24 Is the Champion Pac	pp. 13–14 [ ]
Unit 08: Tactile Algebra: Area Models, Po Kit: Algebra Area Tiles & Factoring Frame	Grades 7–8 (SC-32)	From Al-Khwarizmi's Physic	pp. 15–16 [ ]
Unit 09: Trigonometry, Vectors & Polar/Re Kit: Unit Circle Trigonometry Volvelle	Grades 9–11 (SC-32)	A Stick in Alexandria: How	pp. 17–18 [ ]
Unit 10: Exponential Growth, Compound Int Kit: Proportional Doubling Towers & Logarit	Grades 10–12 (SC-32)	The Scottish Baron Who Dou	pp. 19–20 [ ]
Unit 11: Combinatorics, Probability & Ser Kit: Probability Dice & 10-Sector Spinner	Grades 11–12 (SC-32)	Sir Francis Galton's 1873	pp. 21–22 [ ]
Unit 12: Digital Logic, Base Conversions Kit: 8-Bit Ten-Frame Register & Place Value	Grades 9–12 (SC-32)	Leibniz's Binary Medallion	pp. 23–24 [ ]

BACK OF BOOK: STUDENT SELF-CHECK ANSWER KEY (PAGE 25) & MASTERY CERTIFICATE (PAGE 26)

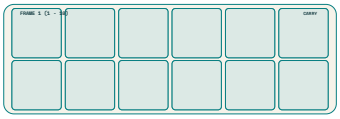
Complete all 12 units and check your work on the SC-6 or SC-32 calculator to earn your Stack Mastery Certificate!

Counting, Cardinality & The One-More Step Counter

Manipulative Kit: Ten-Frame Tray & Counting Pegs • Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Counting, Subitizing & Place Value

STORY OPENER: How Ten Fingers & an Empty Shell Created Place Value & The 1-to-10 Staircase

Long before paper or keyboards existed, ancient shepherds counted flocks with clay pebbles ('calculi' in Latin—the root of 'calculator'!) and Maya astronomers drew a shell for zero so an empty tray had its own symbol. Dropping 1 counter at a time into the Ten-Frame while pressing [1] [+] builds the exact 1-to-10 Staircase from Chapter 3 of From Blocks to the Stack.



KEY BLOCK & STACK RULES: HOW COUNTING MAPS TO THE CALCULATOR

- Every peg placed on the frame matches one count on the calculator screen.
- An empty ten-frame represents zero [0]. Press [C] to start a fresh count.
- To step count by 1: press [1] [ENTER] [1] [+], then press [1] [+] to see the count advance!
- Numbers show quantity (cardinality): the last number said is the total number of pegs.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Place 1 peg into the first well of the Ten-Frame. Have students observe the single object. Question: When we place 1 more peg into the next well, how many pegs will be on our frame? (Prediction: 2)

Press These Keys in Order: [1] [ENTER] [1] [+] [1] [+]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[1]	1	First peg placed: 1 enters register X
[ENTER]	1	1 copied to register Y: holds the current count
[1]	1	Second peg placed: 1 enters register X
[+]	2	Press [+]: 1 + 1 = 2! Screen verifies 2 pegs on board
[1]	1	Third peg placed in frame: 1 enters X
[+]	3	Press [+]: 2 + 1 = 3! Frame and screen match perfectly

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Place 4 pegs on your Ten-Frame. Add 1 more peg. Write the keystrokes to show 4 + 1 on the calculator.

Your Prediction: _____ Calculator Check: [4] [ENTER] [1] [+]

Warm-Up B: Row 1 has 6 pegs and Row 2 has 2 pegs. How many more pegs in Row 1? Write the keystrokes.

Your Prediction: _____ Calculator Check: [6] [ENTER] [2] [-]

Unit 01 Lab: Counting, Cardinality & The One-More Step Counter

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Place 3 pegs on your frame. Add 2 more. Write the keystrokes to find the total.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: A frame has 5 pegs. If you remove 2 pegs, how many remain?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: Clear the calculator by pressing [C]. What number represents an empty ten-frame?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: Step count by ones starting at 1: [1] [ENTER] [1] [+]. Press [1] [+] two more times. What is X?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

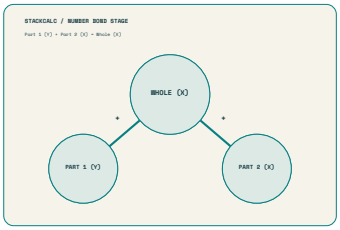
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Part-Part-Whole & Doubles: The Math Domino Studio

Manipulative Kit: Math Domino Studio (12-Frame Tray) + Domino Tiles • Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Number Bonds, Ch. 8: Missing Addend Mysteries & Appendix B: Station F.2

STORY OPENER: Spotting Number Partners at a Glance: Subitizing Domino Pips & Friends of Ten

Your brain has a built-in superpower called subitizing (from the Latin 'subitus', meaning 'sudden'): you can recognize 1 to 5 domino pips instantly without counting one by one! In From Blocks to the Stack, the left half of a domino sits on Shelf Y (Part 1), the right half sits on Shelf X (Part 2), and pressing [+] snaps both halves together into the Whole.



KEY BLOCK & STACK RULES: HOW DOMINOES FIT THE STACK

- Part 1 lives in register Y. Part 2 enters register X.
- Pressing [+] combines both parts into the Whole in register X.
- To find a missing part: start with Whole in Y, enter Part 1 in X, and press [-]!

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Place the [4|3] domino in the 12-frame tray. Count 4 pips on the left (Part 1) and 3 pips on the right (Part 2).
Question: Combine both parts on the tray: 4 + 3 = 7 total pips.

Press These Keys in Order: [4] [ENTER] [3] [+]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[4]	4	Part 1 (4) enters register X
[ENTER]	4	4 copied up to register Y; represents Part 1
[3]	3	Part 2 (3) enters register X
[+]	7	Plus combines Part 1 (Y) + Part 2 (X) = Whole (7) in X!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Pick the [5|4] domino tile. Place it on your 12-frame tray to find the whole. Write the keystrokes.

Your Prediction: _____ Calculator Check: [5] [ENTER] [4] [+]

Warm-Up B: The whole is 8. Part 1 is 5. Find the missing Part 2 using the calculator.

Your Prediction: _____ Calculator Check: [8] [ENTER] [5] [-]

Unit 02 Lab: Part-Part-Whole & Doubles: The Math Domino Studio

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Make 6: Part 1 is 2. What is Part 2? Write the keystrokes.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: Part 1 is 7, Part 2 is 2. What is the Whole?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: Three parts make a whole: 2, 3, and 4. Add them on the stack.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: The Whole is 10. Part 1 is 7. What is Part 2?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

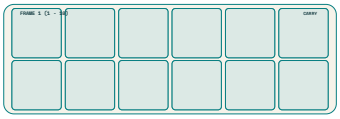
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Ten-Frames: Complements to 10 & 5-Structure

Manipulative Kit: Two 10-Frame Trays (2x5 each, 10 slots) + 3x1 Main Storage Bin (12 slots) + 20 Unnumbered Dots + 2 Double-Height '10' Carry Blocks • Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 3: Making a Ten First & Trading 10 Ones

STORY OPENER: Making a Ten First (8 + 5 = 10 + 3 = 13) & The Double-Height '10' Carry Block Trade

Why is 10 + 3 so much easier for our brains than 8 + 5? Because base-10 place value groups everything into full tens! In Chapter 3 of From Blocks to the Stack, students slide 2 dots from the right tray into the 2 empty pockets of the left tray to complete a full 10, then physically swap all 10 loose dots for one double-height '10' Carry Block.



KEY BLOCK & STACK RULES: TEN-FRAME 5-STRUCTURE & CARRY RULES

- The 2x5 grid holds unnumbered counter blocks. Spatial position builds visual subitizing without numeral distraction.
- Subitizing: 7 is seen immediately as 5 on top plus 2 on bottom.
- Dual 10-Frame Trays: Students use two dedicated 10-frame trays (strictly 10 slots each, solid bottoms) side-by-side on the desk. The 12-slot main storage bin keeps 10-bundles and the 2 carry blocks separate.
- Bundling & Carry: When all 10 slots fill, trade the 10 loose counters for one double-height '10' carry block from the Carry Bay.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Place 7 unnumbered counters into the Ten-Frame: fill the top row of 5, then place 2 in the bottom row. Question: Look at the empty wells without counting one-by-one. 3 empty spaces mean 7 + 3 = 10.

Press These Keys in Order: [1] [0] [ENTER] [7] [-]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[1]	1	Enter 1
[0]	10	Full frame capacity (10) in X
[ENTER]	10	10 lifted to Y
[7]	7	Occupied counters (7) in X
[-]	3	10 - 7 = 3: Exactly 3 empty wells to make 10!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: A ten-frame has 6 counters. How many more counters are needed to make 10?

Your Prediction: _____ Calculator Check: [1] [0] [ENTER] [6] [-]

Warm-Up B: Use the Make-Ten strategy to solve 9 + 6 on your calculator.

Your Prediction: _____ Calculator Check: [9] [ENTER] [1] [+] [5] [+]

Unit 03 Lab: Ten-Frames: Complements to 10 & 5-Structure

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: What is the complement to 10 for 4? Write the keystrokes.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: Solve 7 + 5 using Make-Ten: first make 10, then add 2.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: A ten-frame has 9 counters. How many to make 10?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: Solve 8 + 6 using Make-Ten on the calculator.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

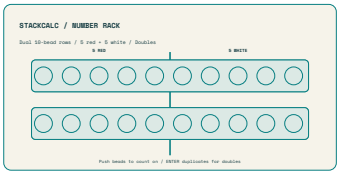
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Number Racks (Rekenrek): 5-Structure & Doubles

Manipulative Kit: Two-Row Number Rack (Rekenrek) • Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 1: Meet Your Math Workbench & Ch. 2: What Happens When You Press [ENTER]?

STORY OPENER: Two Hands, One Workbench: Why 5+5 Bead Rows Mirror the Stack & Instant [ENTER] [+] Doubles

Designed in the Netherlands by math educator Adrian Treffers, the Rekenrek ('calculating rack') splits each row into 5 red and 5 white beads—mirroring the 5 fingers on your left and right hands. In Chapters 1 & 2 of From Blocks to the Stack, pressing [ENTER] clones Shelf X onto Shelf Y just like matching the bottom bead row to the top row for instant doubles!



KEY BLOCK & STACK RULES: NUMBER RACK & ENTER DUPLICATE RULES

- Each row has 5 red beads and 5 white beads.
- Pushing beads to the left counts them. 5 red + 5 white = 10 per row.
- Doubles shortcut: Enter a number, press [ENTER] to duplicate, then press [+]!

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Push 6 beads on the top row (5 red, 1 white). Push 6 beads on the bottom row. Question: Observe the matching rows. 6 + 6 = (5 + 5) + (1 + 1) = 10 + 2 = 12.

Press These Keys in Order: [6] [ENTER] [+]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[6]	6	Top row quantity (6) enters register X
[ENTER]	6	ENTER duplicates 6 into register Y! Both rows now on stack
[+]	12	Plus combines top and bottom rows: 6 + 6 = 12!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Push 7 beads on each row of the Number Rack. Use [ENTER] [+] to double it on the calculator.

Your Prediction: _____ Calculator Check: [7] [ENTER] [+]

Warm-Up B: Use the near-doubles strategy to solve 6 + 7 on the calculator.

Your Prediction: _____ Calculator Check: [6] [ENTER] [+] [1] [+]

Unit 04 Lab: Number Racks (Rekenrek): 5-Structure & Doubles

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Calculate double 8 using [ENTER] [+].

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: Solve near-double 8 + 9: double 8 then add 1.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: Calculate double 9 using [ENTER] [+].

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: Solve 5 + 6 using doubles plus one.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

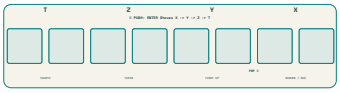
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The 4-Register Stack: Four Operations & Automatic Lift

Manipulative Kit: Place Value Blocks & 4-Register Stack • Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 2: The Number Elevator & Ch. 4: Young Gauss's 10-Second Shortcut

STORY OPENER: Inside the 4-Floor Number Elevator (T, Z, Y, X) & Young Gauss's 10-Second Shortcut

In 1786, 9-year-old Carl Friedrich Gauss stunned his teacher by adding 1 through 100 in seconds—folding the list into 50 pairs of 101! In Chapters 2 & 4 of From Blocks to the Stack, students discover how the 4-Floor Number Elevator (Shelves X, Y, Z, T) automatically lifts finished answers up to Shelf Y when you type the next number, solving multi-step stories with zero parentheses.



KEY BLOCK & STACK RULES: 4-REGISTER STACK ANATOMY

- Four registers: T (top), Z (third), Y (second/operand), X (screen/bottom).
- ENTER duplicates X into Y and pushes older values up into Z and T.
- Two-number operators (+, -, ×, ÷) combine Y and X, dropping Z to Y and T to Z.
- Automatic Lift: Entering a new number after an operation automatically pushes the result to Y!

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Use Place Value unit blocks to represent quantities alongside the 4-line stack display. Question: Problem: (8 - 3) * 2. Predict: 8 - 3 = 5, then 5 * 2 = 10.

Press These Keys in Order: [8] [ENTER] [3] [-] [2] [×]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[8]	8	8 in X
[ENTER]	8	8 copied to Y
[3]	3	3 entered into X
[-]	5	8 - 3 = 5 in X
[2]	2	Automatic lift! 5 lifts to Y; 2 enters X
[×]	10	5 * 2 = 10 in X!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: A baker has 15 cookies, sells 6, and packs the rest into 3 bags. How many per bag? Keystrokes for (15 - 6) / 3.

Your Prediction: _____ Calculator Check: [1] [5] [ENTER] [6] [-] [3] [÷]

Warm-Up B: Calculate (24 / 6) + 15 on the calculator without pressing ENTER twice.

Your Prediction: _____ Calculator Check: [2] [4] [ENTER] [6] [÷] [1] [5] [+]

Unit 05 Lab: The 4-Register Stack: Four Operations & Automatic Lift

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Calculate (14 - 5) * 3 on the stack.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: Calculate 30 / (2 + 3): add denominator first!

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: Calculate (9 + 7) / (8 - 4) using 4 registers.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: A farmer collects 20 eggs, breaks 2, and divides among 6 cartons.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

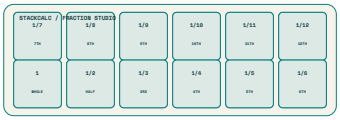
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Fraction Studio: Height Equivalence & Two-Dot Notation

Manipulative Kit: 84 mm Fraction Towers & Twin Comparator Well • Companion Book: From Blocks to the Stack (SC-6 Manual) — Ch. 5: Fractions, Decimals & Ratios (The Pharaoh's Bakery & 84mm Comparator)

STORY OPENER: The Pharaoh's Bakery, Pythagoras's Guitar Strings & Why Your Fraction Towers Stand 84.0 mm Tall

Four thousand years ago, Egyptian scribes had to divide 3 loaves of bread fairly among 4 workers—and instead of cutting every loaf into quarters, they gave each worker 1/2 + 1/4! In Chapter 5 of From Blocks to the Stack, students discover why the Whole block is engineered to 84.0 mm (LCM of 1,2,3,4,6,8,12 -> 84 mm), letting 42 mm (1/2) + 21 mm (1/4) stack flush with three 21 mm (3/4) blocks.



KEY BLOCK & STACK RULES: TWO-DOT FRACTION ENTRY & TOWER EQUIVALENCE

- One Whole tower stands 84.0 mm tall, dividing cleanly into halves (42 mm), thirds (28 mm), fourths (21 mm), sixths (14 mm), eighths (10.5 mm), and twelfths (7 mm).
- Twin Comparator Well: Stack equivalent fraction towers side-by-side against the 84.0 mm Whole tower to verify equivalence by sight and touch.
- Two-dot entry format: [WHOLE] [.] [NUMERATOR] [.] [DENOMINATOR] (or [a ⌧/c] on SC-6).
- Proper fractions have 0 whole: press [.] [3] [.] [4] for 3/4.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Place three 1/4 blocks (3 * 21 = 63 mm). Beside them, stack two Whole blocks and one 1/2 block (84 + 84 + 42 = 210 mm).
Question: Combine heights: 63 mm + 210 mm = 273 mm. In wholes: 3 1/4 (273 / 84 = 3.25).

Press These Keys in Order: [•] [3] [•] [4] [ENTER] [2] [•] [1] [•] [2] [+]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[•]	0.	Leading dot indicates 0 wholes
[3]	0.3	Numerator = 3
[•]	0.3.	Second dot sets fraction entry mode
[4]	3/4	Denominator = 4: displays 3/4 (or 0.75)
[ENTER]	3/4	3/4 lifted to register Y
[2]	2	Whole = 2
[•]	2.	Dot
[1]	2.1	Numerator = 1
[•]	2.1.	Dot
[2]	2 1/2	Denominator = 2: displays 2 1/2
[+]	3 1/4	Plus combines: 3 1/4 (decimal 3.25)!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Build 1/2 and 1/4 with blocks. Verify their combined height against three 1/4 blocks, then check on the calculator.

Your Prediction: _____ Calculator Check: [•] [1] [•] [2] [ENTER] [•] [1] [•] [4] [+]

Warm-Up B: Calculate 2/3 * 3/8 using fraction blocks and verify on the calculator. What block height matches the product?

Your Prediction: _____ Calculator Check: [•] [2] [•] [3] [ENTER] [•] [3] [•] [8] [×]

Unit 06 Lab: Fraction Studio: Height Equivalence & Two-Dot Notation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Add 1/3 + 1/6 using fraction blocks and verify on the calculator.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: Calculate 1 1/2 - 3/4. What block height corresponds to this?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: Compare 3/8 (31.5 mm) to 1/2 (42 mm). What is the decimal equivalent of 3/8?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: Multiply 2/3 * 3/4. Check against your fraction blocks.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

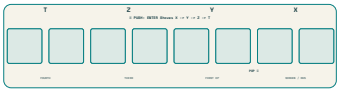
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Postfix Evaluation & Chained Operations: The 4-Register Stack

Manipulative Kit: Counting & Array Tiles + 4-Register Stack • Companion Book: From Blocks to the Stack (SC-6 Ch. 4: Why 24 Is the Champion Packing Number) & Thinking in the Stack (SC-32 Ch. 2: Dijkstra's Railway Siding)

STORY OPENER: Why 24 Is the Champion Packing Number, The Typesetter's Bargain & Dijkstra's 1961 Railway Siding

In the 1500s, typesetters invented parentheses only to squash 2D fraction bars onto flat lines of lead type—then in 1924 Jan Łukasiewicz proved that placing operators after their numbers eliminates every parenthesis, and in 1961 Edsger Dijkstra modeled the 4-register stack as a railway shunting yard! Alongside the 24-Tile Array Studio (where 24 packs into 1×24, 2×12, 3×8, and 4×6 rectangles), students evaluate nested expressions from the inside out.



KEY BLOCK & STACK RULES: POSTFIX EVALUATION & STACK REGISTER RULES

- In postfix evaluation, operands are entered first, followed immediately by operators.
- Evaluation order matches mathematical precedence: evaluate inner sub-expressions first.
- RPN keystroke sequence evaluates complex expressions naturally without parenthesis keys.
- Zero parentheses are needed because stack order guarantees unambiguous evaluation.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-6

Hands-On Setup: Arrange Counting & Array tiles to model the two sub-expressions: (6 + 1) = 7 rows by (9 - 5) = 4 columns. Question: Left sub-expression: 6 + 1 = 7. Right sub-expression: 9 - 5 = 4. Combined 7 x 4 array: 7 * 4 = 28.

Press These Keys in Order: [6] [ENTER] [1] [+] [9] [ENTER] [5] [-] [×]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[6]	6	Value a (6) placed into Register X
[ENTER]	6	ENTER lifts 6 into Register Y
[1]	1	Value b (1) placed into Register X
[+]	7	Left operator (+): Combines Y(6) + X(1) = 7 in X
[9]	9	Value c (9) entered into X; 7 lifted to Y
[ENTER]	9	ENTER lifts 9 into Z and 7 into T
[5]	5	Value d (5) placed into Register X
[-]	4	Right operator (-): Combines Z/Y(9) - X(5) = 4 in X; 7 drops t
[×]	28	Root operator (*): Combines Y(7) * X(4) = 28 in X!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Evaluate (8 + 2) * (14 - 9) by solving the inner quantities first, then checking on the stack.

Your Prediction: _____ Calculator Check: [8] [ENTER] [2] [+] [1] [4] [ENTER] [9] [-] [×]

Warm-Up B: Evaluate (15 - 5) / (3 + 2) by computing the numerator and denominator, then verifying on the stack.

Your Prediction: _____ Calculator Check: [1] [5] [ENTER] [5] [-] [3] [ENTER] [2] [+] [÷]

Unit 07 Lab: Postfix Evaluation & Chained Operations: The 4-Register Stack

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Evaluate (4 + 3) * (10 - 6) and check on the stack.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 2: Solve (20 - 4) / (2 * 4) using inner-first evaluation.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 3: Evaluate (18 - 6) / (1 + 2) on the 4-level stack.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

Investigation 4: Evaluate 5 + (3 * 4) by computing the inner product first.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-6 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

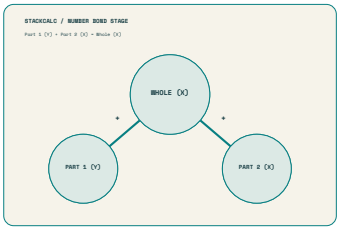
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Tactile Algebra: Area Models, Polynomials & Quadratic Factoring

Manipulative Kit: Algebra Area Tiles & Factoring Frame • Companion Book: From Blocks to the Stack (SC-6 Ch. 4: Difference of Two Squares) & Thinking in the Stack (SC-32 Ch. 7: The Duel in Milan & [POLY])

STORY OPENER: From Al-Khwarizmi's Physical Squares to the 1535 Duel in Milan & Kahan's [POLY] Solver

In 820 AD in Baghdad, Al-Khwarizmi solved $x^2 + 10x = 39$ by literally arranging physical square and rectangular tiles to 'complete the square'—and in 1535 in Milan, Niccolò Tartaglia defeated Antonio Fiore in a 30-problem public algebra duel overnight! Students build physical $(x + a)(x + b)$ tile rectangles and verify roots with the SC-32 [POLY] solver.



KEY BLOCK & STACK RULES: AREA MODELING & QUADRATIC FACTORING RULES

- Polynomial tiles represent algebraic terms: large square = x^2, rectangle = x, small square = 1.
- An algebraic expression factors when tiles form a solid rectangle with dimensions $(x + a)$ by $(x + b)$.
- Zero pairs: positive $(+x)$ and negative $(-x)$ tiles cancel out to zero to complete rectangular arrays.
- SC-32 quadratic solver [POLY]: enter coefficients a, b, c to display roots and verify factors on the 4-line LCD!

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-32

Hands-On Setup: Place 1 x^2 tile, 5 x-tiles, and 6 unit-1 squares on the factoring stage. Question: Arrange into a solid rectangle: width is $(x + 2)$, length is $(x + 3)$. Total Area = $x^2 + 5x + 6$.

Press These Keys in Order: [1] [ENTER] [5] [ENTER] [6] [↗] [POLY]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[1]	1	Coefficient a = 1 entered into X
[ENTER]	1	1 lifted to Y
[5]	5	Coefficient b = 5 entered into X
[ENTER]	5	b in Y, a in Z
[6]	6	Coefficient c = 6 entered into X
[↗] [POLY]	-2	Press [⊠] [POLY]: Root 1 = -2 in Y (factor: x + 2), Root 2 = -

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Arrange tiles for $x^2 + 7x + 12$ into a rectangle to find its factors, then verify with [POLY].

Your Prediction: _____ Calculator Check: [1] [ENTER] [7] [ENTER] [1] [2] [↗] [POLY]

Warm-Up B: Factor $x^2 - 5x + 6$ using zero pairs on the Algebra Stage and verify on the calculator.

Your Prediction: _____ Calculator Check: [1] [ENTER] [5] [+/-] [ENTER] [6] [↗] [POLY]

Unit 08 Lab: Tactile Algebra: Area Models, Polynomials & Quadratic Factoring

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Factor $x^2 + 6x + 8$ into a rectangle of algebra tiles, then verify with [POLY].

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 2: Factor $x^2 - 4$ (difference of two squares) using zero-pair x-tiles.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 3: Factor $x^2 + 5x + 4$ using polynomial tiles and verify.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 4: Factor $x^2 - 5x + 6$ using zero pairs.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

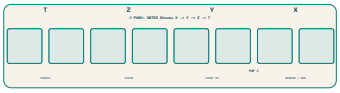
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Trigonometry, Vectors & Polar/Rectangular Coordinates

Manipulative Kit: Unit Circle Trigonometry Volvelle • Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 4: Trigonometry, Vectors & Polar Coordinates (A Stick in Alexandria)

STORY OPENER: A Stick in Alexandria: How Eratosthenes Measured the Earth with a Shadow & Crosswind Vectors

In 240 BC, Eratosthenes measured a 7.2° shadow angle in Alexandria at noon on the summer solstice while a well in Syene 800 km away had zero shadow—calculating Earth's circumference within 1% using pure geometry! With the 3-disc Unit Circle Volvelle and [->theta,r] polar conversion, students resolve right triangles and aircraft crosswind vectors in a single keystroke.



KEY BLOCK & STACK RULES: VECTOR DECOMPOSITION & POLAR CONVERSION RULES

- A 2D vector v = <x, y> has horizontal run x in Y and vertical rise y in X.
- Rectangular to Polar [⊠] [->theta,r]: Converts (x, y) into angle theta (in X) and magnitude r (in Y).
- Trig functions [SIN], [COS], [TAN] operate on the angle in register X.
- To swap between angle and magnitude after conversion, press [x swap y]!

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-32

Hands-On Setup: Align the Unit Circle Volvelle cursor to horizontal run 4 and vertical rise 3 (forming a right triangle with hypotenuse 5). Question: Run = 4, Rise = 3. Predict magnitude: r = sqrt(4² + 3²) = 5. Predict angle: arctan(3/4) ≈ 36.87°.

Press These Keys in Order: [4] [ENTER] [3] [↶] [▸θ,r] [x⇄y]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[4]	4	Enter horizontal component x = 4
[ENTER]	4	4 lifted to register Y
[3]	3	Enter vertical component y = 3 in X
[↶] [▸θ,r]	36.87	Rectangular to Polar! Screen displays angle theta = 36.87°
[x⇄y]	5.00	Swap registers: Y displays vector magnitude r = 5.00!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: A surveyor walks 60 m East and 80 m North. Find the straight-line displacement magnitude and heading.

Your Prediction: _____ Calculator Check: [6] [0] [ENTER] [8] [0] [↶] [▸θ,r] [x⇄y]

Warm-Up B: Two cables pull on a post: Vector A is <5, 2> and Vector B is <-2, 4>. Find the resultant vector magnitude.

Your Prediction: _____ Calculator Check: [5] [ENTER] [2] [+/-] [+] [2] [ENTER] [4] [+] [↶] [▸θ,r] [x⇄y]

Unit 09 Lab: Trigonometry, Vectors & Polar/Rectangular Coordinates

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Find magnitude and angle for vector <12, 5>.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 2: Find the hypotenuse of a right triangle with legs 7 and 24.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 3: Calculate sin(30°) on the calculator.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 4: A vector has magnitude 10 at angle 45°. What are its x and y components?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

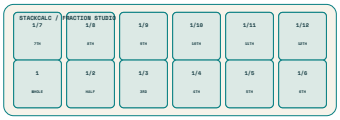
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Exponential Growth, Compound Interest & Logarithms

Manipulative Kit: Proportional Doubling Towers & Logarithmic Blocks • Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 3: Transcendentals (John Napier & Euler) & Ch. 11: Time Value of Money

STORY OPENER: The Scottish Baron Who Doubled the Lifespan of Astronomers (John Napier), Euler's e & The Rule of 72

In 1614, John Napier at Merchiston Castle realized that logarithms turn tedious multiplication into simple height addition—the exact engineering principle of our Prime Factor blocks (where h = 15 * log2(p) mm)! Paired with Jakob Bernoulli's continuous compounding limit e and Fibonacci's Liber Abaci (1202), students model exponential doubling and half-life decay.



KEY BLOCK & STACK RULES: EXPONENTIALS, CONTINUOUS COMPOUNDING & DOUBLIN

- Exponential growth formula: A = P * e^(r*t) for continuous growth, or A = P * (1 + r/n)^(n*t).
- Natural exponential [e^x] raises Euler's constant e to power X.
- Natural logarithm [LN] inverts exponential growth: ln(e^x) = x.
- Doubling time formula: t = ln(2) / r. Stack evaluates ln(2) without rounding loss.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-32

Hands-On Setup: Stack Fraction Blocks representing exponential doubling: 12 mm, 24 mm, 48 mm, 96 mm. Question: Invest $1,000 at 7% continuous interest for 10 years: A = 1000 * e^(0.07 * 10) = 1000 * e^0.7. Will it double? (e^0.7 ≈ 2.01).

Press These Keys in Order: [0] [•] [0] [7] [ENTER] [1] [0] [×] [e^x] [1] [0] [0] [0] [×]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[0] [•] [0] [7]	0.07	Rate r = 0.07 entered in X
[ENTER]	0.07	Rate lifted to Y
[1] [0]	10	Time t = 10 entered in X
[×]	0.7	Exponent r * t = 0.70
[e^x]	2.0138	e^0.70 = 2.01375 (Growth factor: more than doubled!)
[1] [0] [0] [0]	1000	Principal P = 1000 (growth factor lifts to Y)
[×]	2013.75	Total accumulated amount A = $2,013.75!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: A bacteria culture starts at 500 and triples every hour. Use [y^x] to calculate population after 4 hours.

Your Prediction: _____ Calculator Check: [3] [ENTER] [4] [y^x] [5] [0] [0] [×]

Warm-Up B: A radioactive isotope has decay constant k = 0.035. Find its half-life: t = ln(2) / k.

Your Prediction: _____ Calculator Check: [2] [LN] [0] [•] [0] [3] [5] [÷]

Unit 10 Lab: Exponential Growth, Compound Interest & Logarithms

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Calculate e^2 on the calculator to 4 decimal places.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 2: Calculate doubling time for an investment growing at 8% continuous interest.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 3: Compute 10^3.5 using [□] [10^x].

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate log10(500) using [□] [LOG].

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

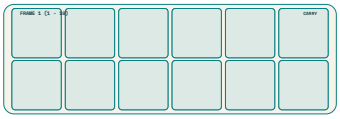
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Combinatorics, Probability & Series Accumulation

Manipulative Kit: Probability Dice & 10-Sector Spinner • Companion Book: From Blocks to the Stack (SC-6 Ch. 7: Galton's Bean Machine) & Thinking in the Stack (SC-32 Ch. 9: Ceres, Guinness & [Sum

STORY OPENER: Sir Francis Galton's 1873 Bean Machine, The Asteroid Ceres & The Guinness Brewery Chemist

In 1873, Sir Francis Galton built a board studded with pins (the Quincunx) to show how independent random bounces settle into a bell-shaped curve—while in 1908 at the Guinness Brewery in Dublin, chemist William Sealy Gosset ('Student') invented small-sample statistics with n - 1 degrees of freedom! Students roll fair 16 mm polyhedral dice and accumulate trials with [SUM+].



KEY BLOCK & STACK RULES: COMBINATIONS, PERMUTATIONS & STACK FACTORIALS

- Factorial [⊠] [x!] multiplies n * (n-1) * ... * 1.
- Permutations (order matters): nPr = n! / (n - r)!.
- Combinations (subsets/groups): nCr = n! / (r! * (n - r)!).
- Stack accumulation: [SUM+] accumulates data points for instant mean and standard deviation.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-32

Hands-On Setup: Use the 4 precision dice and 10-sector spinner to model multi-stage discrete outcomes. Question: How many ways to choose a committee of 3 from 8 candidates? 8C3 = 8! / (3! * 5!) = 40320 / (6 * 120) = 56.

Press These Keys in Order: [8] [↶] [x!] [3] [↶] [x!] [÷] [5] [↶] [x!] [÷]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[8]	**8**	Total candidates n = 8
[↶] [x!]	**40320**	8! = 40,320 in register X
[3]	**3**	Subset r = 3 (40320 lifts to Y)
[↶] [x!]	**6**	3! = 6 in X
[÷]	**6720**	40320 / 6 = 6720 in X
[5]	**5**	Remaining n - r = 5 (6720 lifts to Y)
[↶] [x!]	**120**	5! = 120 in X
[÷]	**56**	6720 / 120 = 56 combinations!

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: A lottery draws 4 winning numbers from a pool of 10. How many combinations? Calculate 10C4.

Your Prediction: _____ Calculator Check: [1] [0] [↶] [x!] [4] [↶] [x!] [÷] [6] [↶] [x!] [÷]

Warm-Up B: Accumulate data points 12, 15, 18, 21 using [SUM+]. What is the sample count n and mean?

Your Prediction: _____ Calculator Check: [1] [2] [Σ+] [1] [5] [Σ+] [1] [8] [Σ+] [2] [1] [Σ+]

Unit 11 Lab: Combinatorics, Probability & Series Accumulation

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Calculate 6! (6 factorial) on the calculator.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 2: How many ways can 5 runners finish 1st, 2nd, and 3rd? (5P3 = 5! / 2!).

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 3: Calculate 7C2 (combinations of 2 from 7).

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 4: Accumulate 5 and 15 with [SUM+]. What is the mean?

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

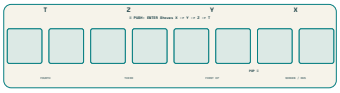
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Digital Logic, Base Conversions & Computer Architecture

Manipulative Kit: 8-Bit Ten-Frame Register & Place Value Blocks • Companion Book: Thinking in the Stack (SC-32 Manual) — Ch. 10: Computer Science, Base-N & 36-Bit Word Logic

STORY OPENER: Leibniz's Binary Medallion, Claude Shannon's 1937 Relay Basement & NASA's 36-Bit IBM 7090 Word

In 1697, Gottfried Wilhelm Leibniz designed a medallion showing that every number can be built from 1 and 0 alone; in 1937, 21-year-old MIT student Claude Shannon proved electrical switches execute Boolean algebra; and in 1961, NASA's IBM 7090 computed orbital trajectories in 36-bit binary words! Using an 8-slot Ten-Frame register and [BASE], students decode bytes, hex nibbles, and bitwise masks.



KEY BLOCK & STACK RULES: NUMBER BASES & BITWISE LOGIC GATES

- Binary (Base 2): digits 0 and 1. Hexadecimal (Base 16): digits 0-9 and A-F.
- Press [⊠] [BASE] to cycle between DEC (decimal), HEX (hexadecimal), OCT (octal), and BIN (binary).
- Bitwise AND: result bit is 1 only if BOTH bits are 1 (used for masking and subnets).
- Bitwise OR: result bit is 1 if EITHER bit is 1. Bitwise XOR: result bit is 1 if bits DIFFER.

GUIDED WALKTHROUGH: BUILD WITH BLOCKS & PRESS KEYS ON YOUR SC-32

Hands-On Setup: Use the 8 wells of the Ten-Frame as an 8-bit byte. Place pegs in wells for active '1' bits, leave empty for '0'.
Question: Place pegs at positions 0, 1, and 3: binary 00001011_2 = 8 + 2 + 1 = 11_10. Predict hex value: 0B_16.

Press These Keys in Order: [1] [1] [↶] [BASE]

Button Pressed	Screen (X)	What Happens Inside the 4-Level Stack (T, Z, Y, X)
[1] [1]	11	Enter decimal integer 11 in DEC mode
[↶] [BASE]	HEX 0B	Press [⊠] [BASE]: screen displays Hexadecimal 0B!
[↶] [BASE]	BIN 1011	Press [⊠] [BASE]: screen displays Binary 1011!
[↶] [BASE]	OCT 13	Press [⊠] [BASE]: screen displays Octal 13!
[↶] [BASE]	DEC 11	Press [⊠] [BASE]: returns to Decimal 11

WARM-UP CHECK: TRY THESE TWO CHALLENGES BEFORE STARTING PAGE 2!

Warm-Up A: Enter decimal 255. Use [⊠] [BASE] to convert to HEX and BIN. What is the binary pattern?

Your Prediction: _____ Calculator Check: [2] [5] [5] [↶] [BASE]

Warm-Up B: Calculate bitwise OR for 12 (1100) and 3 (0011). Predict the binary and decimal result.

Your Prediction: _____ Calculator Check: [1] [2] [ENTER] [3] [OR]

Unit 12 Lab: Digital Logic, Base Conversions & Computer Architecture

Name: _____ Date: _____ Score: _____ / 4

DIRECTIONS: (1) Build with blocks & sketch in Box 1. (2) Show your math strategy in Box 2.
(3) Write your Prediction BEFORE pressing keys, then check on your StackCalc calculator!

Investigation 1: Convert decimal 42 to binary using [⊠] [BASE].

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 2: Convert hexadecimal 2A to decimal.

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 3: Evaluate bitwise AND: 15 (1111) AND 7 (0111).

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

Investigation 4: Evaluate bitwise XOR: 12 (1100) XOR 10 (1010).

1. Block Model / Diagram Sketch:

2. Written Strategy & Stack Plan:

3. My Prediction: _____

Final Verified Answer: _____

[ ] Verified on SC-32 Stack

EXIT REFLECTION: What did Register Y hold during Investigation 4 while you typed your second number?

_____

Student Self-Check Answer Key

Check your Warm-Up and Investigation answers after completing each unit. Full step-by-step traces are in Book 1.

Unit 01: Counting, Cardinality & The One- (Pre-K – Kind) — Warm-Up A: 5 pegs | B: 4 more pegs

Investigation 1: 5 • **Investigation 2: 3**
Investigation 3: 0 (Zero) • **Investigation 4: 4**

Unit 02: Part-Part-Whole & Doubles (Grades K–1) — Warm-Up A: 9 pips | B: Missing Part is 3

Investigation 1: 4 • **Investigation 2: 9**
Investigation 3: 9 • **Investigation 4: 3**

Unit 03: Ten-Frames (Grade 1) — Warm-Up A: 4 counters | B: 15

Investigation 1: 6 • **Investigation 2: 12**
Investigation 3: 1 • **Investigation 4: 14**

Unit 04: Number Racks (Rekenrek) (Grades 1–2) — Warm-Up A: 14 beads | B: 13

Investigation 1: 16 • **Investigation 2: 17**
Investigation 3: 18 • **Investigation 4: 11**

Unit 05: The 4-Register Stack (Grades 3–4) — Warm-Up A: 3 cookies per bag | B: 19

Investigation 1: 27 • **Investigation 2: 6**
Investigation 3: 4 • **Investigation 4: 3 eggs**

Unit 06: Fraction Studio (Grades 4–5) — Warm-Up A: 3/4 (height 63 mm) | B: 1/4 (height 21 mm)

Investigation 1: 1/2 (42 mm) • **Investigation 2: 3/4 (63 mm)**
Investigation 3: 0.375 • **Investigation 4: 1/2 (42 mm)**

Unit 07: Postfix Evaluation & Chained Ope (Grades 5–6) — Warm-Up A: 50 | B: 2

Investigation 1: 28 • **Investigation 2: 2**
Investigation 3: 4 • **Investigation 4: 17**

Unit 08: Tactile Algebra (Grades 7–8) — Warm-Up A: (x + 3)(x + 4), roots -3 a | B: (x - 2)(x - 3), roots 2 an

Investigation 1: (x + 2)(x + 4); roots -2, -4 • **Investigation 2: (x - 2)(x + 2); roots 2, -2**
Investigation 3: (x + 1)(x + 4); roots -1, -4 • **Investigation 4: (x - 2)(x - 3); roots 2, 3**

Unit 09: Trigonometry, Vectors & Polar/Re (Grades 9–11) — Warm-Up A: Displacement: 100 m, Headi | B: Magnitude: 6

Investigation 1: theta = 22.62°, r = 13 • **Investigation 2: 25**
Investigation 3: 0.5 • **Investigation 4: x = 7.07, y = 7.07**

Unit 10: Exponential Growth, Compound Int (Grades 10–12) — Warm-Up A: 40,500 bacteria | B: 19

Investigation 1: 7.3891 • **Investigation 2: 8.66 years**
Investigation 3: 3162.28 • **Investigation 4: 2.6990**

Unit 11: Combinatorics, Probability & Ser (Grades 11–12) — Warm-Up A: 210 combinations | B: Count n = 4, Mean x-bar = 16

Investigation 1: 720 • **Investigation 2: 60**
Investigation 3: 21 • **Investigation 4: Mean = 10**

Unit 12: Digital Logic, Base Conversions (Grades 9–12) — Warm-Up A: HEX: FF, BIN: 11111111 (8 | B: Binary: 1111, Decimal: 15

Investigation 1: 101010 (32 + 8 + 2) • **Investigation 2: 42**
Investigation 3: 7 (0111) • **Investigation 4: 6 (0110)**

Certificate of Hands-On Stack Mastery

Awarded for completing Units 01–12 of the StackCalc Hands-On Math Lab Workbook.

CERTIFICATE OF HANDS-ON MATHEMATICS MASTERY

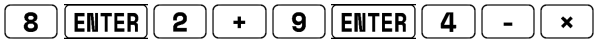
This certifies that: _____

has completed the 12 Hands-On Math Lab investigations from Counting & Number Bonds through Fractions, Algebra Area Models, Trigonometry, Logarithms, Probability, and Digital Logic—building physical block models first and verifying every solution on the 4-level StackCalc RPN stack (T, Z, Y, X).

[ ] Elementary Stack Explorer (Units 01–07 • SC-6) **[ ] Scientific Stack Master (Units 08–12 • SC-32)**

Teacher / Parent Signature: _____ Date: _____

ESSENTIAL SC-6 & SC-32 KEYCAP SEQUENCES TO REMEMBER

Doubling Any Number in Register X (e.g., 2 × 7 = 14)	
Parenthesis-Free Chained Math: (8 + 2) × (9 - 4) = 50	
Two-Dot Fraction Addition: 3/4 + 1/2 = 1 1/4	
Quadratic Roots of x² - 5x + 6 = 0 (Roots: 2, 3)	
Pythagorean Hypotenuse of 3 and 4 -> 5	